

DBMS Quick Guide — RAG Test Document

1. Database and DBMS

A database is an organized collection of related data. A Database Management System (DBMS) is software used to create, store, retrieve, update, and manage data in databases. Examples include MySQL, PostgreSQL, Oracle Database, and Microsoft SQL Server.

2. Relational Database

A relational database stores data in tables. A table consists of rows and columns. Each row represents a record, while each column represents an attribute. Tables can be related to one another using keys.

3. Keys

A primary key uniquely identifies each row in a table and cannot contain NULL values. A candidate key is an attribute or set of attributes that can uniquely identify a row. A foreign key is an attribute that references a key in another table and is used to establish relationships.

4. Normalization

Normalization is a database design technique used to reduce redundancy and avoid update anomalies. First Normal Form (1NF) requires atomic values and no repeating groups. Second Normal Form (2NF) requires 1NF and removal of partial dependency on a composite key. Third Normal Form (3NF) requires 2NF and removal of transitive dependency.

5. SQL

SQL is used to interact with relational databases. SELECT retrieves data, INSERT adds rows, UPDATE modifies existing rows, and DELETE removes rows. The WHERE clause filters rows, GROUP BY groups rows for aggregation, and ORDER BY sorts the result.

6. Transactions

A transaction is a logical unit of database work. The ACID properties are Atomicity, Consistency, Isolation, and Durability. Atomicity means a transaction is completed fully or not at all. Consistency ensures that database rules remain satisfied. Isolation controls how concurrent transactions interact. Durability means committed changes survive system failures.

7. Indexing

An index is a data structure that can speed up data retrieval. Indexes are useful for columns frequently used in search conditions, joins, and sorting. However, indexes require additional storage and can make INSERT,

UPDATE, and DELETE operations more expensive.

Practice Questions for RAG

1. What is a DBMS?
2. What is the difference between a primary key and a foreign key?
3. Explain 1NF, 2NF, and 3NF.
4. What does ACID stand for?
5. What does the WHERE clause do in SQL?
6. What are the advantages and disadvantages of indexes?